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REVIEW ARTICLE

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Equations to describe the amount and rate of sorption

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Abstract

The partition of materials that react with soil between the solid and the solution phase, and how this changes with time, can often be described by a simple equation: $S = a\ c^{b1}t^{b2}$ where S is the amount sorbed, c is the solution concentration, t the time of contact, and a, b_1 and b_2 are parameters. However, when the range of values for sorption is large, it is apparent that both b_1 and b_2 increase with decreasing sorption. At low values for sorption, b_1 approaches 1, and sorption plots are nearly linear. These observations are consistent with a mechanistic model in which it is postulated that the materials react with heterogenous sites. As the amount of sorption decreases, the heterogeneity of the occupied sites decreases. This is why b_1 increases. Because there is heterogeneity of occupied sites, there is a range of rates for the subsequent reaction. This is why the rates are proportional to a fractional index of time. It is better to describe the effects of time this way than by using several first-order equations.

KEYWORDS

cadmium, modelling, phosphate, rate of reaction, selenite, sorption plots

1 | INTRODUCTION

Ions that react with soil may be divided into two categories. One category contains those ions for which the bonds are electrostatic. It includes sodium or calcium ions. For this category, reaction depends on the charge of the ions and how closely they can approach the surface. This article is about the other category: ions which form a specific bond with the surface particles. This category includes anions such as phosphate, arsenate, selenite and fluoride, and hydrolysable cations such as copper and cadmium. It is important to understand how these ions are partitioned between the solution and the solid phase; the proportion in the solution, and how it changes through time, are important characteristics, strongly influencing both the rate of movement to plant roots and the propensity to be lost in drainage.

It was proposed by Kuo and Lotse (1973) that the behaviour of phosphate can be summarised by a simple equation which can be represented as follows:

$$S = a\,c^{b1}t^{b2} \tag{1}$$

where S is the amount sorbed, c is the solution concentration, t is the time of contact, and a, b_1 and b_2 are parameters. This equation has subsequently been used by other authors to describe the effects of concentration and of time on the sorption of many specifically sorbed ions. In contrast, the effects of time have also been described by reaction schemes involving sets of first-order reactions (Elbana et al., 2018; Selim et al., 1992; Selim & Zhang, 2013; Zhang & Selim, 2007).

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In this paper, I discuss the conditions under which Equation (1) applies. I show that whether it is appropriate depends on the range over which sorption is measured. I also show that equations based on it describe the effects of time more efficiently than equations based on multiple first-order reactions. Finally, I show that these effects of time are consistent with a mechanistic model and are consequences of reaction with heterogenous surfaces.

Highlights

- The changes with time in the concentration of ions in the soil solution is important.
- In many, but not all, situations a simple equation can describe this.
- This equation is consistent with a mechanistic model.
- It is better to use the simple equation than to postulate many first-order reactions.

2 | METHODS

I used data from published papers, some from papers published by others, some from my own publications. I apologise that this retrospective re-evaluation inevitably involves some self-citation.

Where it was necessary to digitise data from images, I used WebPlotDigitizer 4.5. (https://automeris.io/WebPlotDigitizer).

When fitting equations to the data, I regarded the equations that related sorption to concentration as simultaneous with $S = (c_i – c)\ Ssr$ where S is the amount sorbed, c_i is the initial concentration, c is the observed concentration and Ssr is the solution soil ratio. I wrote a computer program which, for a given initial concentration and for given values of the parameters, found the value of concentration (c) for which the two equations were true. The program then found the values of the parameters that minimised the sums of squares for the predicted and observed values of log concentration. It, therefore, used the variables controlled by the experimenter, the initial concentration and the soil solution ratio, to predict the observed variable, concentration. This process is described in detail by Barrow (2008).

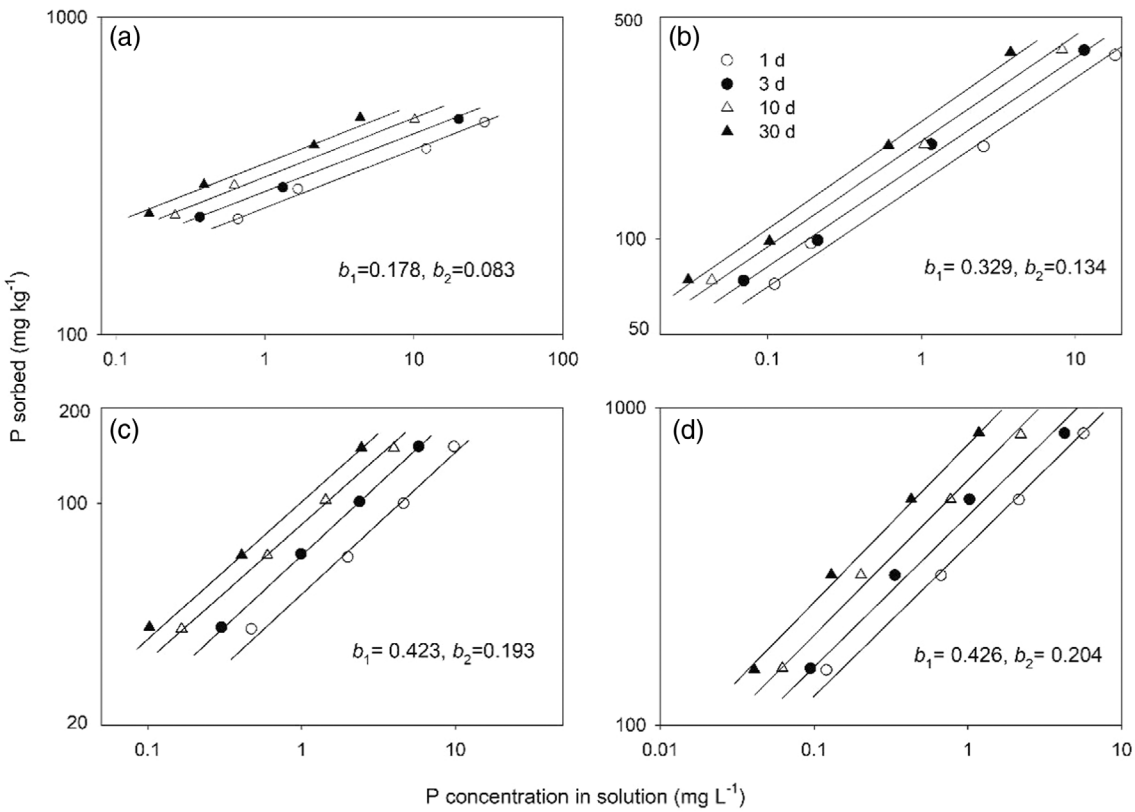

FIGURE 1 Phosphorus sorption plots for four different soils. Soil (a) is a kaolinitic subsoil from Western Australia. The other soils are topsoils from (b) Chainat, Thailand, (c) Sutherlin, California, and (d) Molokai, Hawaii. For each plot, the horizontal scale comprises a 1000-all fold range of concentrations and the vertical scale a 10-fold range values for sorption. The slopes can therefore be compared. The lines indicate the fit of the Equation (2): $S = a\ c^{b1}t^{b2}$ and the values of the parameters b_1 and b_2 are shown. (Drawn from the data of Barrow (1980)).

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I compared the observations with the output of my mechanistic model (Barrow, 1983, 2015). The model is based on the proposition that specifically sorbed ions react with heterogenous variable-charge surfaces, and that the initial adsorption reaction is followed by solid-state diffusion into the reacting particles. I used the

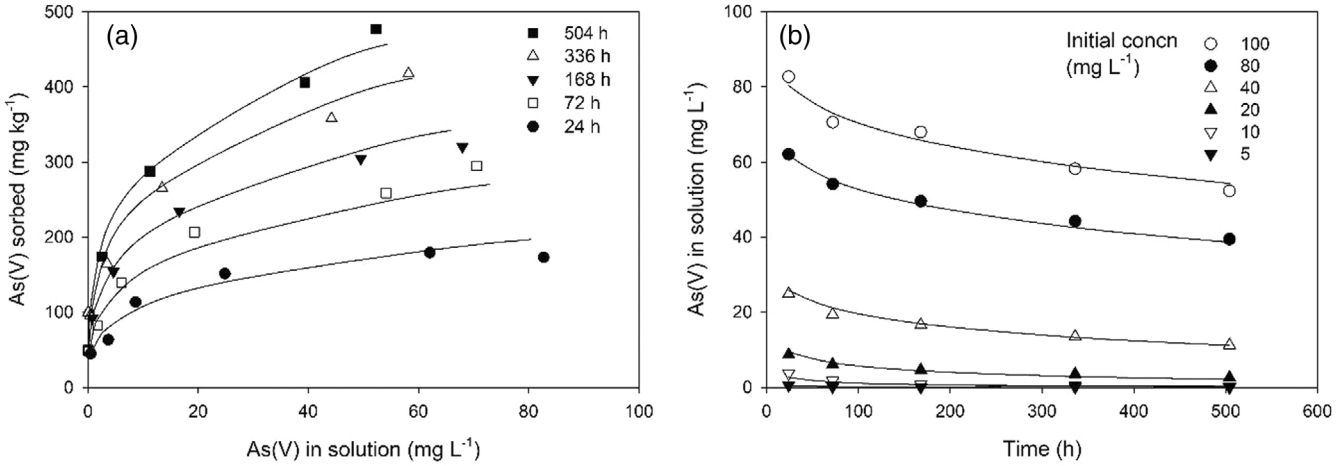

FIGURE 2 Effect of initial arsenate concentration, and period of reaction on arsenate sorption. Points are for Olivier soil of Zhang and Selim (2005); the lines are from the fit of Equation (1): $S = a\ c^{b1}t^{b2}$ where S is the amount sorbed, c is the solution concentration, t the time of contact, and a, b_1 and b_2 are parameters. The values were: a, 20.36; b_1, 0.290; b_2, 0.314; R^2, 0.998.

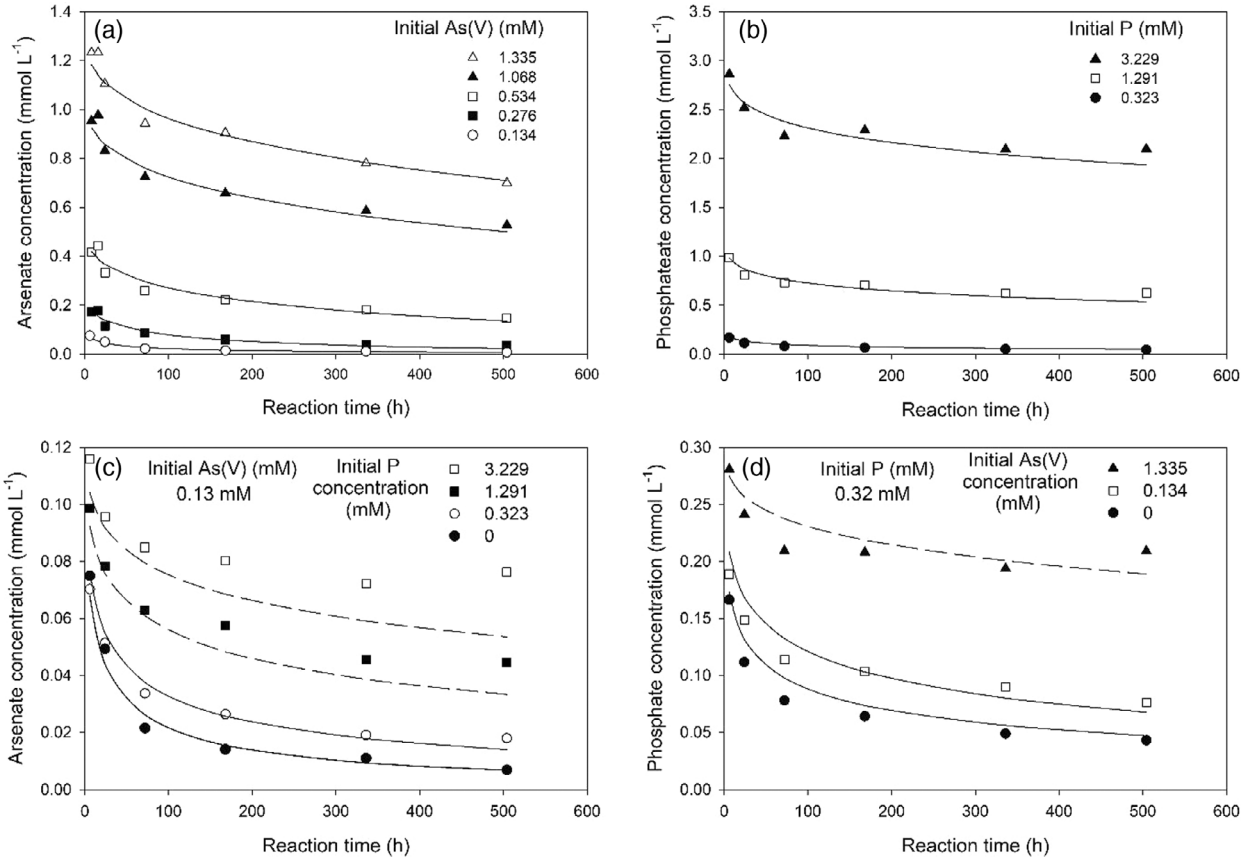

FIGURE 3 Effect of initial arsenate and phosphate concentrations, and period of reaction on arsenate and phosphate sorption. Parts a and b show individual effects; parts c and d show competition. In parts c and d, broken lines indicate extrapolations because values for the "opposite" concentration were not available. Points are for Olivier soil of Zhang and Selim (2005); the lines are from the fit of Equation (3) $S = a\ c_1t^{b2}/(c_1 + d\ c_2)^{(1\text{-}b)}$ where S is the amount sorbed, c is the solution concentration, t the time of contact, and a, b_1, b_2 and d are parameters. The values were as follows (P, As): a, 1.96, 0.921; b_1, 0.417, 0.350; b_2, 0.260, 0.338; d, 2.36, 0.163; R^2, 0.992, 0.995.

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program DIFFPLUS of Barrow (1987). The program divides the distribution of electric potentials into 30 slices. Within each slice, a modified Langmuir equation is used to calculate sorption. The modification is that concentration is replaced by a surface activity function (a_{is}) where

$$a_{is} = c\,\alpha_i \exp(-z_i\,\psi\,F/\mathrm{RT}) \tag{2}$$

α_i is the fraction present as the reacting ion, z_i is its valence, ψ is the electric potential of the midpoint of the slice, F is the Faraday, R is the gas constant, and T is the temperature (K).

I ran the model over a 10^9-fold range in concentrations and recorded modelled sorption after 1, 3, 10, and 30 days. The values of the parameters were as follows: maximum adsorption, 3000 μmol^{-1} m^2, midpoint of the normal distribution of potentials, −70 mV, standard deviation 50 mV, binding constant, 0.1 L μmol^{-1}, fraction dissociated (α_i), 0.1 d^{-1}, first feedback term, 125 mV μmol^{-1} m^2, second feedback term, 25 mV μmol^{-1} m^2. These parameters are appropriate for phosphate sorption at pH approximately 6. The effects of varying the values of some of the parameters were demonstrated by Barrow (1987).

3 | RESULTS

3.1 | The observations

When sorption is plotted against solution concentration Equation (1) produces a series of parallel lines (Figure 1); the slope of these lines reflects the value of the parameter b_1 and the distance apart reflects the value of b_2. Figure 1 illustrates some of the range in the values of these two parameters. With a wider selection of soils, a much greater range in values is observed. I showed (Barrow, 2020) that, subject to two qualifications, this equation applied to phosphate sorption by 95 different soils with b_1 ranging from 0.07 to 0.69, and b_2 from 0.076 to 0.43. One of the qualifications was that many soils contained significant levels of phosphate and the equation had to be modified by adding an intercept term. The other qualification was that for soils with pH_{H20} of 8 or above, the relationship only held until the precipitation of calcium phosphate started.

Nevertheless, others have preferred a different approach to describe the rate of reaction. For example, Zhang and Selim (2005) postulated that arsenate reaction with soils involved separate sites which they specified as equilibrium, kinetic and irreversible. Transfer between such sites involved first-order reactions; in this case, they specified four first-order reaction coefficients. Figure 2 shows the results for Olivier soil. To describe the data in Figure 2a, they tabulate separate values for each of the five periods of reaction for the parameters a and b_1 of Equation (1)—that is, 10 parameters. To describe the data in Figure 2b, they tabulate four parameters for their "three-phase reversible-irreversible" model. Figure 2 shows that the data are closely described by Equation (1), involving just three parameters.

Zhang and Selim (2007) modified their approach, adding another kind of proposed site. Their modified model involves sites that are described as adsorbed, equilibrium, kinetic, consecutive irreversible, and concurrent irreversible. Their modified model involves five first-order reaction rates. They applied their model to sorption of phosphate, and of arsenate and for competition between them. When competition is involved, Barrow et al. (2005) showed that Equation (1) can be modified to:

$$S = a\,c_1 t^{b2}/(c_1 + d\,c_2)^{(1-b1)} \tag{3}$$

where c_1 indicates the concentration of one reactant, c_2 is the concentration of the other reactant, and d is the competition coefficient. When $c_2 = 0$, this equation reverts to Equation (1). Zhang and Selim (2007) used three parameters for phosphate and three for arsenate to describe sorption and competition at 24 h, plus five separate parameters for each reactant to describe the rate of reaction. Figure 3 shows that the data can be well described with three parameters for each reactant plus two competition parameters. The competition parameters are not

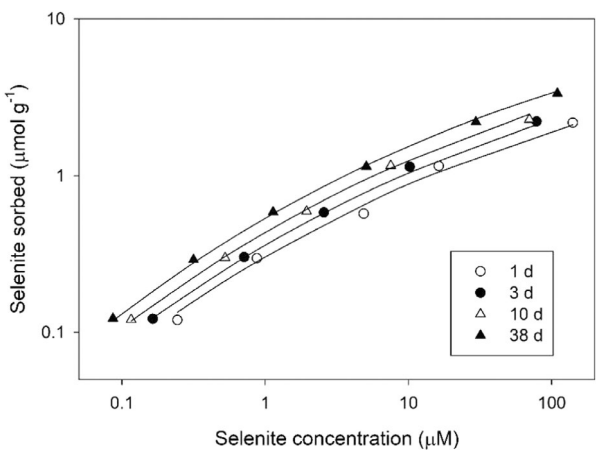

FIGURE 4 Effects of selenite concentration and time of reaction on selenite sorption by a Western Australian topsoil. The points are derived from the data of Barrow and Whelan (1989); the lines are derived from fitting a modified form of Equation (2). The modified equation may be written:

$$S = a\,c^{b1} t^{b2} \quad \text{where} \quad b_1 = \beta_1 c^{\beta 2}.$$

The values of the parameters are: a, 0.303; β_1, 0.529; β_2, -0.0606; b_2, 0.157; R^2, 0.997.

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symmetrical; one is not the reciprocal of the other as they would be if the simple chemical competition was involved.

It is apparent from the above that phosphate and arsenate sorption by soils are well described by Equation (1). Phosphate is a macro nutrient with a high affinity for sorption sites in soils. It is often studied at high solution concentrations and gives rise to high values for sorption. In contrast, selenite is studied at lower concentrations and lower values for sorption. For selenite, plots of sorption against the concentration on a log–log scale are not linear but gently curved (Figure 4). That is, the value of parameter b_1 of Equation (2) varies with concentration. This can be described by writing.

$b_1 = \beta_1 c^{\beta 2}$. The modified equation has four parameters instead of three. Figure 4 shows that the modified equation gives a good description of the observations. As for phosphate and arsenate, the effects of the time of reaction are described as proportional to a fractional power of time.

Sorption of phosphate and selenite appears to involve the same reaction sites. If the concentration of selenite is adjusted to allow for its weaker sorption, Figure 5 shows that the sorption curves become coincident. The combined plot now extends over a 10,000-fold range of concentrations and the slope term (b_1) ranges from approximately one at low concentration to approximately 0.25 at high concentration (Figure 5).

Sorption of trace elements and of pollutants is also often studied at lower concentrations. The data of Selim et al. (1992) is especially valuable because there is a 100,000-fold range of solution Cd concentration (Figure 6a). Selim et al. (1992) applied a Freundlich equation to describe these results. That is, they assume that b_1 was constant. Figure 6a shows that on a log–log scale, plots of sorption against concentration were not linear but significantly curved. That curvature was also closely described by writing $b_1 = \beta_1\ c^{\beta 2}$. Further, on a log–log scale, the plots were not parallel. There was a highly significant improvement by adding the following term: $b_2 = \beta_3\ c^{\beta 4}$.

Thus, the data, including the time trends, were closely described by an equation with five parameters (Figure 6b). Selim et al. (1992) describe these time trends

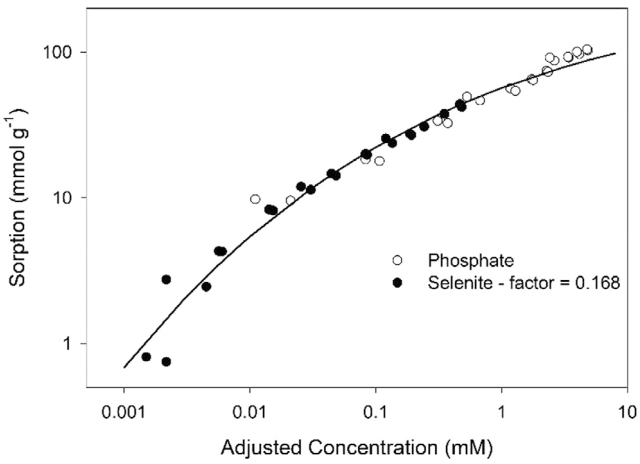

FIGURE 5 Comparison of sorption of phosphate and selenite in a Chilean Andisol. For selenite, the concentration has been multiplied by the factor shown (0.168). Redrawn from the data of Barrow et al. (2005). For low values of concentration and of sorption, the value of the slope term (b_1) is approximately unity; for high values of concentration and sorption it is approximately 0.25.

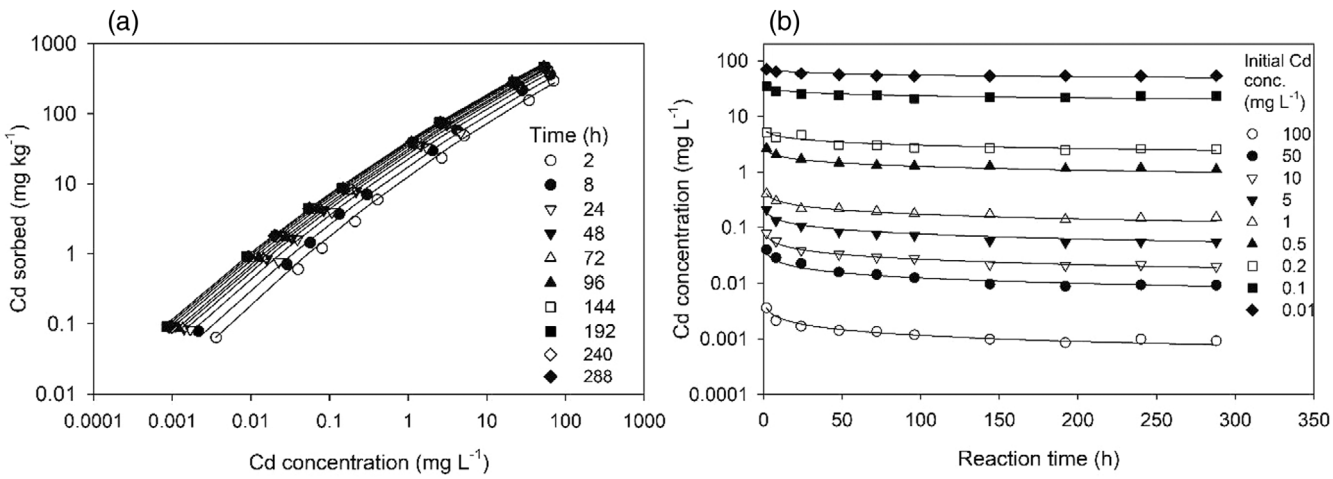

FIGURE 6 Effect of time and of initial cadmium concentration on sorption of cadmium by Windsor soil. Data points are from Selim et al. (1992). The lines represent the fitted equation relating sorption (S) to solution concentration (c): $S = a\ c^{b1} t^{b2}$, where $b_1 = \beta_1\ c^{\beta 2}$. and $b_2 = \beta_3\ c^{\beta 4}$. the fitted values for the parameters were: a, 10.92 ; β_1, 0.819 ; β_2,—0.0275; β_3, 0.231 ; β_4, −0.0821; R^2, 0.999. The value of b_1 (the Freundlich index term) ranges from 0.99 for $c = 0.001$, to 0.72 for $c = 100$.

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using 45 parameters: 5 parameters for each initial cadmium concentration. These parameters were regarded as representing two reversible and one irreversible mechanism.

Figures 5 and 6 show that at low solution concentrations the value of b_1 approaches unity. That is, sorption plots are close to linear. This has not always been appreciated. Elbana et al. (2018) studied five heavy metals and concluded that sorption plots were non-linear. However, McLaren et al. (1983) pointed out that copper sorption studies published prior to their paper had used copper concentrations well in excess of those found under normal soil conditions. They quoted other publications that showed that, at low concentrations appropriate to those of soil solutions, sorption plots were linear. Some of their linear sorption plots are shown in Figure 7.

In summary, Equation 2 applies in many situations and especially for studies on phosphate, but it is not universal. When studied over a wide concentration range, neither b_1 nor b_2 is constant. At low concentrations, b_1 approaches unity and sorption plots are close to linear. Its value decreases with increasing sorption. The value of b_2 also decreases with increasing sorption.

3.2 | Explanation

These seemingly disparate observations are reproduced by the mechanistic model (Figure 8). Over a large range of concentrations, the curvature is clear, but over a small range, it is not marked. Figure 9 shows the results for concentrations ranging from 0.1 to 100 mmol L^{-1}, but

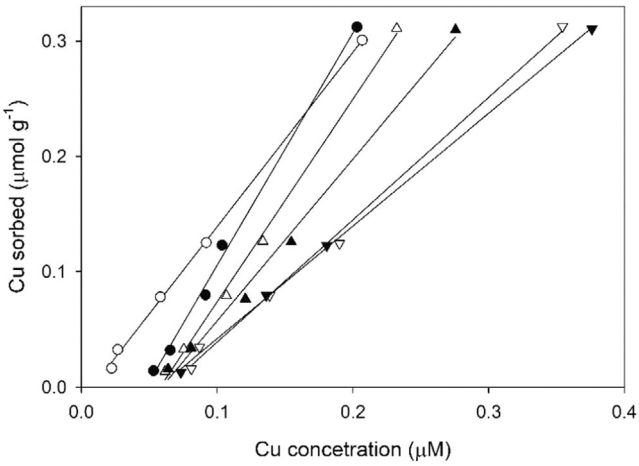

FIGURE 7 Sorption of copper at low solution copper concentrations. (Drawn from the data of McLaren et al. (1983).)

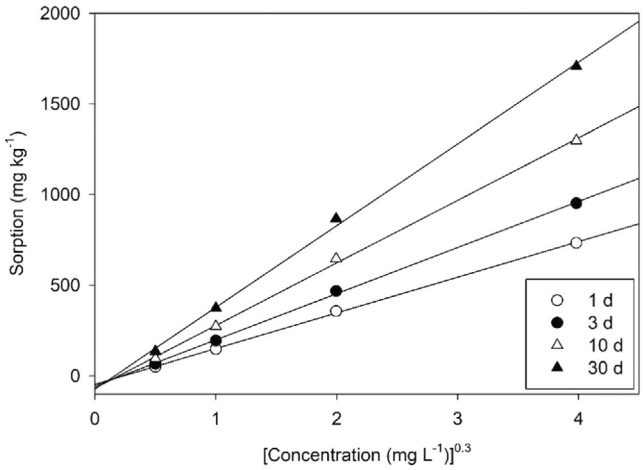

FIGURE 9 Replot of some of the data from Figure 8. The values for concentration are those ranging from 0.1 to 100 mg L^{-1}, but plotted on a scale of concentration raised to a fractional power.

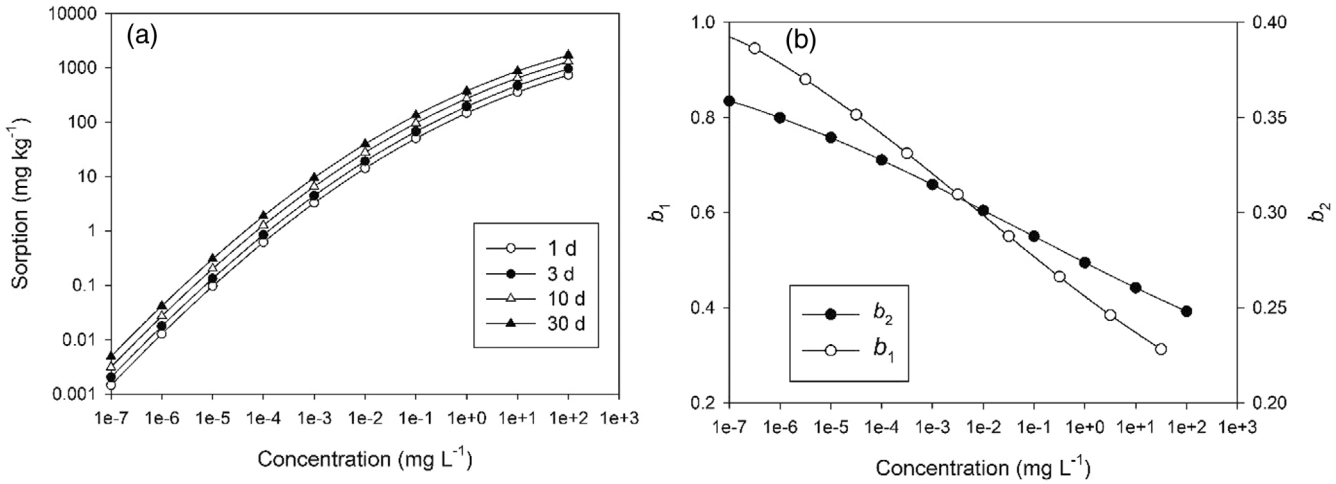

FIGURE 8 Output from the model "DIFFPLUS" of Barrow (1987) described by Barrow (1983, 2015) to model the effects of initial concentration and of period of reaction on sorption.

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plotted on a scale of concentration raised to a fractional power—a plotting device often used to inspect such data. The deviations are small. Extrapolation to the vertical axis gives a negative intercept. Such extrapolations are sometimes used to assess phosphate status, and this shows there is an inherent error.

At low values for concentration, and for sorption, the value of b_1 approaches 1 (Figure 8b). Under these conditions, only the most favourable sites are occupied; there is little heterogeneity in the occupied sites. Further, the equation describing the sorption on these sites may be written as follows:

$$S = M\,k_i\,a_{is}/(1 + k_i\,a_{is}) \tag{4}$$

where M is the maximum sorption. When concentration, and therefore a_{is}, are small, the denominator approaches unity and sorption is linearly related to a_{is}. Linear sorption plots at low concentrations therefore occur.

As the concentration is increased, increasingly less-favourable sites are occupied; the heterogeneity of occupied sites therefore increases. The decreasing value for b_1 reflects this increased heterogeneity.

The surface concentration is the driving concentration for the reaction that follows adsorption. There is a range of surface concentrations, and it follows there will be a range of rates for the subsequent reaction. This is why the rate of reaction is proportional to a fractional power of time. There is much evidence that this reaction is diffusive penetration (Barrow, 2022). As the concentration is increased, and the heterogeneity of occupied sites increases, the value of b_2 increases. The effect is small and that is why it is only noticed when there is a large range of concentrations, as in Figure 6.

4 | DISCUSSION

This work shows that, for a given soil, the values of b_1 and b_2 depend on the amount of sorption and therefore on the concentration and identity of the reactant. It is convenient to summarise here the other factors that determine the value of these characteristics. An important factor is the inherent heterogeneity of soils. This is illustrated in Figure 1, and is modelled by varying the standard deviation of the distribution of potentials (Barrow, 2020). The greater the heterogeneity, the larger the values. It may be relevant that much research work has been done on highly-weathered topsoils, especially in Australia. These soils may have a smaller range of reacting materials, low heterogeneity, and high values for these parameters.

There are two other factors that may affect the values. One is the pH. As the pH is increased, the electric potential of the components may change in a different way, giving rise to a different distribution. For phosphate, I found (Barrow, 1983) that, as the pH increased the log–log sorption plots became steeper (b_1 increased). For specifically sorbed cations, large effects of pH on the amount of sorption occur. The observation by Elbana et al. (2018) that, for zinc, nickel and cadmium, b_1 decreased with increasing pH may be largely due to increased sorption.

The other factor is fertiliser history. The reaction of phosphate with a soil would be expected to change the distribution of electric potentials. Figure 10 shows that after prolonged reaction with phosphate not only is the reaction with further additions of phosphate decreased, but the slope of the log–log sorption plots is also increased, suggesting a greater spread of occupied sites.

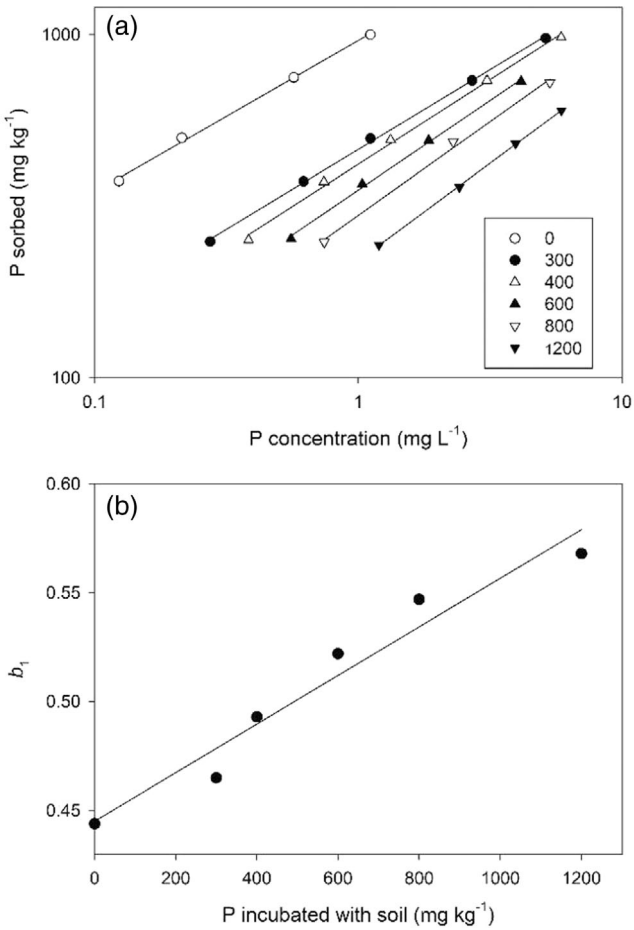

FIGURE 10 Effect of previous additions of phosphate on the sorption of subsequent additions. Phosphate had been added to a soil at the levels indicated (mg kg^{-1}) and incubated at 25°C for 12 months. Further sorption was then measured by mixing samples of the soil with phosphate solutions for 24 h at 25°C. Part a shows the upper part of the sorption plots; part b shows the slope of these plots (b_1) against the levels of P initially added. (Drawn from the data of Barrow (1974)).

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The value of b_2 is partly dependent on the value of b_1 because both reflect heterogeneity. Under restricted conditions, such as those of Figure 1, the two characteristics are correlated. However, the value of b_2 is also influenced by other factors. One is the nature of the particles involved in the reaction. They would be expected to differ in the extent to which they are prone to diffusive penetration. Another is the phosphate status including fertiliser history. As the phosphate status of soil is increased, diffusive penetration is decreased (Barrow & Debnath, 2014). However, the biggest influence is the laboratory treatment; almost always, samples of soil are mixed vigourously with solutions containing the reactant. It is difficult to understand why; it is scarcely because the reaction is so rapid that vigourous mixing is needed to ensure uniformity. The problem is that vigourous mixing causes mutual abrasion of particles thus exposing new surfaces and causing an overestimation of the amount and rate of sorption (Barrow & Shaw, 1979). Soils differ in the extent to which they are prone to this problem making it difficult to compare soils.

5 | CONCLUSIONS

Equation (1) can be regarded as a base equation. It is useful under common but restricted conditions. It can, and should, be modified especially when there is a large range of concentrations.

Thus, the effects of concentration can be efficiently described by functions in which concentration is raised to a fractional power. Similarly, the effects of time can be described using functions in which time is raised to a fractional power. Both effects are consequences of heterogeneity and are therefore both mechanistically based. It is better to describe the effects of time this way than by postulating a sequence of first-order reactions involving sites for which there is no independent evidence.

AUTHOR CONTRIBUTIONS

N. J. Barrow: Conceptualization; investigation; writing – review and editing; software; methodology; writing – original draft; project administration; formal analysis.

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DATA AVAILABILITY STATEMENT

Data sharing not applicable to this article as no datasets were generated or analysed during the current study.

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